PROPOSAL OF
AE *Innoflex* THREE LAYER CO EXTRUSION BLOWN FILM LINE 1950_55*75*55
FOR
M/s. KANHA POLYFAB AND POLYMERS, INDORE

Ref. No. AEP/DOM/1950/001
Date: 04/10/2025

(ADROIT EXTRUSION SCOPE OF SUPPLY)

ITEM #	ITEM DESCRIPTION
1	Three Nos. Extruders of 55/75/55 mm screw diameter and L/D ratio of 32:1, Imported Bimetallic screw & barrel, Ceramic Heater, 40/100/40 HP AC Motor & AC variable frequency Drive. Three Candle type Manual Screen Changers for 55/65/55 mm Extruders.
2	Complete extrusion controls on main panel with Cold start protection and with Touch Panel.
3	One Nickel plated Three Layer Spiral Mandrel Die with lip diameter of 400 mm, complete with die adapters and carriage.
4	Advanced Saturn design Air Ring Package consisting of highly efficient air ring, distributor manifold, High Pressure 20 HP Blower with AC Frequency drive.
5	One Bubble Stabilizing Basket with Silicon roller cage. Motorized Open Close and Up Down operation.
6	One 360-degree rotation bottom supported Horizontal Oscillating Haul Off. Collapsing frame with Segmented PBT Roller, side guides, Main Nip with 3 HP AC Drive. Linespeed is 80 MPM.
7	Aluminum Idler rollers as per layout drawing.
8	One Secondary nip with edge slitting assembly and edge trimming assembly.
9	Two Separate Surface Winders of 1850 mm film width. Manual roll change over mechanism, digital length counter, 04 nos.- 3" Air shaft, bow roller, 02 HP AC Motor and Drive. Tension Control through Loadcell.
10	Tower Structure to support and mount Bubble stabilizing Basket, Collapsing Frame, Oscillating Haul Off, Secondary Nip, Web aligner, Corona Treater etc.
11	

- **BASIC PRICE EX-WORKS, UNPACKED ……. INR 1,47,50,000/-**

  (INR ONE CRORE FOURTY SEVEN LAC FIFTY THOUSAND ONLY)

  FINAL PRICE AFTER DISCOUNT……. INR 1,17,50,000/-

ELECTRICALS

Panels	: Powder coated panels with exhaust fans.
Synchronisation	: (1) For all extruders (2) For Nips & winder drives.
Temp. controllers	: Multispan / Omron.
Drives	: Danfoss or equivalent variable frequency drives.
Hauloff	: PLC based operation.

TOWER STRUCTURE

Tower structure with idler rollers to support and mount bubble cage, collapsing frame, Haul-off, Secondary nip etc. Beams will be provided at Bubble Cage for lifting the die parts.

OPTIONAL EQUIPMENTS

Digital Edge Guide Adroit make: Rs. 2,50,000/-
Corona Treater Jain Electrotech make: Rs. 6,75,000/-
Air cooled Air Chillers for Airring: Rs. 11,40,000/-
Air cooled Water Chiller for Gearbox and Feed Throat: Rs. 3,00,000/-
Vertical Granule Mixer with Dryer 200 kg Adroit make: Rs. 2,25,000/-
Hopper Loader: Rs. 1,65,000/-

INDICATIVE PERFORMANCE

Product to be made	: High Quality Three Layer Film
Max Pumping Capacity	: 340-360 KG / HR
Max Output * on 400 mm die 75% LL + 25% LD	: 230 – 270 KG / HR
Thickness Variation	: +/- 8% above 40 μ and +/- 10% upto 40 μ or +/- 4 micron whichever is higher Film thickness over 90% periphery.
Raw Materials	: LDPE, LLDPE, HDPE, mLLDPE, PP etc.
Thickness Range	: 20 – 200 micron
Lay flat Width	: 1850 MM maximum process able width.

Provided that the temperature of air at the air ring will not be more than 10 – 12° C and 2.5 BUR+ Widths only. Ambient temperature and relative humidity within normal range.

LIMITATIONS:

Although we can design our equipment to as high a standard as possible, there are many factors other than equipment that can significantly affect the quality of the film. As these factors are out of our control, they are not covered by our performance guarantee. These factors include but are not limited to:

- Resin – Bad Batches, Off-spec. Resins, Changes in additives or the use of resins other than Specified can produce poor film.
- Interfacial Instability – Mismatched viscosity in adjacent layers, can be reduced by matching resins or increasing thickness of thin layers.
- Blocking – Weather effects, resins, additives and processing conditions can induce blocking, even with a tower of recommended height running the guarantee structures.
- Gels – Defective resins, dirty screws or processing conditions can cause gels.
- Gloss and Haze – Can be affected by resin, temperature profiles or inappropriate purge resins.
- Melt fracture due to a too narrow die gap and material properties. Processing aid may be required.
- Curling – Asymmetrical film structures will curl.
- Film Characteristics – Impact resistance, tear strength and other film qualities can be radically affected by processing conditions, resins and additives.

EXTRUDER - 55 MM (02 NOS.)

Screw Diameter	: 55 mm single screw extruder mounted on a sturdy frame.
L/D ratio	: 32:1
Type	: Barrier
Screw Speed	: 115 RPM
Barrel	: Water cooled grooved feed section
Material	: Bimetallic
Heating System	: Ceramic Band type Heaters (AUM or equivalent)
No. of Zones	: 04 Nos. on barrel
Hopper	: MS fabricated with glass window for visual inspection. Magnetic grill provided.
Main Drive	: 40 HP AC motor (ABB) with frequency variable drive.
Transmission System	: Motor directly coupled with helical gearbox. (Zeal)
Screen changer	: Candle type

EXTRUDER 75 MM (01 NO.)

Screw Diameter	: 75 mm single screw extruder mounted on a sturdy frame.
L/D ratio	: 32:1
Type	: Barrier
Screw Speed	: 115 RPM
Barrel	: Water cooled semi grooved feed section
Material	: Bimetallic
Heating System	: Ceramic Band type Heaters (Hitco or equivalent)
No. of Zones	: 05 Nos. on barrel
Hopper	: MS fabricated with glass window for visual inspection. Magnetic grill provided.
Main Drive	: 100 HP AC motor with frequency variable drive.
Transmission System	: Motor directly coupled with helical gearbox.
Screen changer	: Candle type

THREE LAYER DIE

Material of Construction	: Hardened high strength alloy steel.
Surface Treatment	: Electroless Nickel plated & highly polished melt paths.
Die Size	: 400 MM diameter with lips of 2.3 mm gap.
Die setting	: Die adjusting bolts will be provided.
Distribution	: Spiral distribution.
Heating Zones	: 04 Nos.
Heating System	: Ceramic band heaters (AUM or equivalent)

BUBBLE CAGE

Type	: Calibration bubble guide basket with 6 arms arranged to provide full support. Bubble contact is through Aluminum roller for minimum drag.
Actuation of arms	: Motorized Open-Close and Up Down operation.
Film Width range	: 1000 to 1750 mm

HORIZONTAL HAULOFF

Construction	: The haul off will be shipped in assembled parts.
Main Nip rollers	: 2 Nos. mounted in bearings. One chrome plated roller and one rubber roller movable pneumatically.
Nip roller width	: 1950 mm
Nip roller drive	: 3 HP Gear motor (Bonvario, Italy) with variable frequency drive.
Collapsing Frames	: Aluminum rollers
Idler rollers	: Adequate quantity as per layout.
Oscillation	: 360 degree with end limit switch sensing and override protection ensuring even thickness variation distribution on rolls giving excellent roll geometry.
Turnbars	: 02 Nos. Aluminum rollers with Hard Anodized mounted in haul off.

SECONDARY NIP & SURFACE WINDERS

Additional Nip	: 2 Nos. mounted in bearings. One chrome plated roller and one rubber roller movable pneumatically.
Edge slit assembly	: Highly efficient design for trouble free operation reduces trim wastage.
Nip roller width	: 1950 mm
Nip roller drive	: 2 HP Gear motor with variable frequency drive.
Tension control	: Through Loadcell.
Surface Winders (01 No.)	: Maximum web width of 1850 mm with Manual Changeover.
Roll diameter	: 700 mm diameter or 400 kg weight in single up Which ever reaches first. Bow roller prior to drum roller for wrinkle free winding.
Surface winder drive	: 2 HP Gear motor with variable frequency drive.
Tension control	: Through Loadcell
Length counter meter	: Provided
Trim Suction Blower	: Provided

ITEMS NOT INCLUDED IN QUOTATION

- Electrical accessories
  1. Main source of electrical power or generator with stabilizer or UPS if needed by you.
  2. Power Cables from Main supply to distribution panel and to individual panels.
  3. Earthing to all equipments to be provided by you. (Recommended 3 pits)
- Finished roll handling system
  1. Unloading from winders.
- Installation of equipment
  1. Customer to arrange contractor or team of mechanical/structural/electrical team to work under supervision of engineers.
- All interconnecting piping for water & air to the equipments.

UTILITIES REQUIRED

- Electrical Power Standard Voltages

Network Structure : TN-S neutral conductor capable of carrying full load, Separate earthing for power circuit and control circuit earthing voltage should be less than 1V
Service Voltage for drive motors : 3 x 415 V, 50 Hz, with N
Service Voltage for heaters : 3 x 415 V, 50 Hz, with N
Control Voltage : 230 V/AC, 50 Hz.
Permissible Voltage Fluctuation : + 5% to - 5%
Permissible Frequency Fluctuation : +/- 3 Hz.

SR. NO.	EQUIPMENTS	QTY.	APP. HEATING LOAD KW	APP. MOTIVE LOAD KW
1	EXTRUDER 55 MM	2	22.4	60
2	EXTRUDER 65 MM	1	17.2	45
3	SCREEN CHANGER	3	6.75	
4	3 LAYER DIE	1	24.30	
5	AIR RING	1		11.5
6	OSC. HAULOFF	1		2.25
7	SURFACE WINDER WITH ADD. NIP	3		4.5
		TOTAL KW	70.65	118.50

WATER

- **Treated water**

Classification : Boiler feed water
pH Value : 7.5 - 8
Operating pressure : 3 - 4 Bar
Permissible Fluctuation : +/- 0.25 Bar
Max. hardness : 5 Degrees dH
Max. non-carbonate hardness : 2 Degrees dH
Filter to max. : 40 microns
General criteria : Free from corrosion substances and any foreign particles.
Water requirement : 30 LPM X 3 extruders = Total 90 LPM at 25 Degree centigrade.

COMPRESSED AIR

Operating pressure : 5 - 7 Bar
Permissible Fluctuations : +/- 0.25 Bar
Dew point : - 25 Degrees C
Dust-oil-water Content : None

SR. NO.	EQUIPMENTS	QTY.	PRESSURE (BAR)	TOTAL AMT. OF AIR M3/HR
1	AT DIE DURING STARTUP & OPERATION	1	5-7 BAR	8.5
2	OSC. HAULOFF	1	5-7 BAR	1.7
3	SURFACE WINDER WITH ADD. NIP	1	5-7 BAR	5.1
			TOTAL	15.3

RECOMMENDED ENVIRONMENT

Allocation of the extrusion plants in rooms as dry and dust free as possible.
Air Temperature
Classification : Boiler feed water
Electrical equipment : Permissible ambient temperature is max. 45 deg. For elect. equipment.
Air Ring Blower inlet : Maximum 40 degrees Centigrade
Air Humidity : Formation of condensed moisture has to be avoided otherwise special measures are necessary.
Altitude : Below 1000 m above sea level.

The above mentioned specifications are minimum requirements to ensure proper operation of Extrusion System. Any deviation affects the performance of the Extrusion system.

TERMS & CONDITIONS

PRICES:
Prices are Un Packed, Ex-works, Ahmedabad, Gujarat basis. All Government taxes and duties are not considered. These are applicable as per rule. Prices are valid for 30 days from the date of quotation.

PACKING & FORWARDING: Packing charges will be charged extra.

INSURANCE: Transit insurance of the machine is to be cover by the buyer. In the event of any Insurance claim, the procedure would be done by the buyer.

TRANSPORTATION: Cost of the transport will be borne by the customer.

DELIVERY: 04 Months from the date receipt of technically and commercially clear purchase order with minimum 40% irrevocable security deposit. Failure to lift the machinery against balance payment even a month after the machine will ready, then ADROIT EXTRUSION has right to sell it to another customer. In that case, you have to wait till another same machine will be ready by us. The agreed prices will change in case buyer fails to take delivery of the ordered machine within 30 days from the date of conveying readiness of the machine to the buyer.

PAYMENT TERMS: We require 40% of the Total value of order as irrevocable security deposit. Balance payment along with all incidental charges, Taxes & Duties are to be paid before delivery. The advance is non-interest bearing. In the event of cancellation of the order, advance will be adjusting against order cancellation charges. Offsetting or retention of payments based on any counterclaims whatsoever of the Buyer are not admissible. Once Machine Dispatched, It cannot be Returned at any Condition.

JURISDICTION: All Contract of sell Shall be Subject to Ahmadabad jurisdiction only.

PRE-DISPATCH TESTING & TRIALS: We follow discrete testing methods. This method involves discrete testing of individual system elements like Winders, Extruders etc. to our pre- designed standards to ensure faultless running of the complete line on installation at Customer end. For the wet trials, the buyer is required to arrange all the input raw materials or Seller will arrange the same at buyer cost.

ERECTION & COMMISSIONING: Our service engineers will supervise erection, installation and commissioning of the machine. Customer will provide skilled and semi skilled labor for the purpose of erection & commissioning. Customer will arrange Erection Team. Customer will ensure that all utilities and power connections are ready before customer calls our service engineers. The customer will pay To & Fro 2 AC Train Fare, Lodging, Boarding, and Transportation during period of the services provided.

Thanking you,

Yours truly,

FOR **ADROIT EXTRUSIO**

Jepaliya

ADROIT EXTRUSION AHMEDABAD

Urveesh Jepaliya

STANDARD WARRANTY TERMS

WARRANTY PERIOD:

All Mechanical components warranty for 1 year, All Pneumatic Parts Warranty for 1 year (Dry Air Compulsory with Air Compressor. If not Dry Air, then Warranty will not Apply), Drives Warranty for 1 year (Voltage stabilizer compulsory, If not then Warranty will not Apply) All Electrical & Electronics parts like Switch, Contactor, MCB, etc. warranty for 06 months from date of commissioning against faulty material or bad workmanship and undertake to repair/replacement the defective parts within period. The defect brought to notice has to be genuine. The cost of the new materials or repairing of the defective ones will be borne by the Seller. However, the cost of transportation and destination clearance charges etc. will be borne by the buyer.

- The buyer is required to return the defective parts or equipment to seller or to the manufacturer and the cost for the same will be bear by the buyer

In case of payment terms agreed upon in the contract are not been obeyed, the warranty expires consequentially.

WARRANTY EXCLUDES FOLLOWING:

Parts made of rubber, plastic, glass. Bearings, wearing parts, fuses, heaters, lamps, contactors, MCB's etc. The warranty does not cover damages caused by dropping, fire, floods or other natural calamity, misuse, accident or improper installation. The life span of screw and barrel depends upon abrasive material being processed and hence cannot be specified.

WARRANTY IS LIMITED to the extend of the repairs or replacement of manufacturing defects and other things or workmanship. No liability is assumed beyond such replacements.

Any condition or other matters relating to this quotation not expressly stimulated will be a matter of mutual discussion and agreement at the time of accepting the order.

CANCELLATION

It is understood that order once placed cannot be cancelled for any reason whatso ever. In case of cancellation of the order, the entire amount of the advance will be forfeited.